

Thermal PDR Document

2.1.4 Thermal Control System Overview

The Thermal Control System (TCS) is responsible for integrating the heat-generation, heat-distribution, and environmental-interface subassemblies into a single system aimed at maintaining electronics, sensors, mechanisms, and structural elements at 20 °C (293 K) during the entire mission. Most avionics, batteries, sensors, and mechanical components operate within this range, and it provides a stable thermal baseline for scientific instruments. The TCS has to maintain this internal temperature while the Martian ground temperature ranges between 128 K and 324 K, and the air temperature varies from 149 K to 272 K, according to requirement TCS-01 to ensure all systems remain at operational readiness.

The temperature fluctuations from Mars' environment means that the TCS will need to be capable of providing heating during nighttime, efficient heat retention by insulation, and heat rejection during the daytime or high dissipation periods. The system was, therefore, designed with a combination of active and passive methods that work together to stabilize the internal temperature.

The heat generation subassembly is the source of the active heating that is used to prevent any part of the hardware from going below the 20 °C level. The system includes patch heaters, thermostats, and solid-state controllers, all of which have redundancies to fulfill the TCS-04 requirement. Each of the patch heaters can generate up to 250 Watts and is glued to batteries, electronic boards, and other anti-thermally sensitive components. Thermostats and solid-state control relays regulate the heaters by monitoring multiple temperature sensors throughout the system. In the case a sensor determines that the temperature of a component or a section of the structure is below 20 °C, the corresponding heater is turned on. When the temperature reaches an adequate range again, the heater is turned off. With this control method, the TCS ensures that the heaters are only used when necessary, thus respecting the power consumption limit prescribed in TCS-06. Since the surface can reach extremely low temperatures during Martian nights (up to 128 K), these heaters are the elements ensuring that other components don't freeze and maintain their operational readiness at all times.

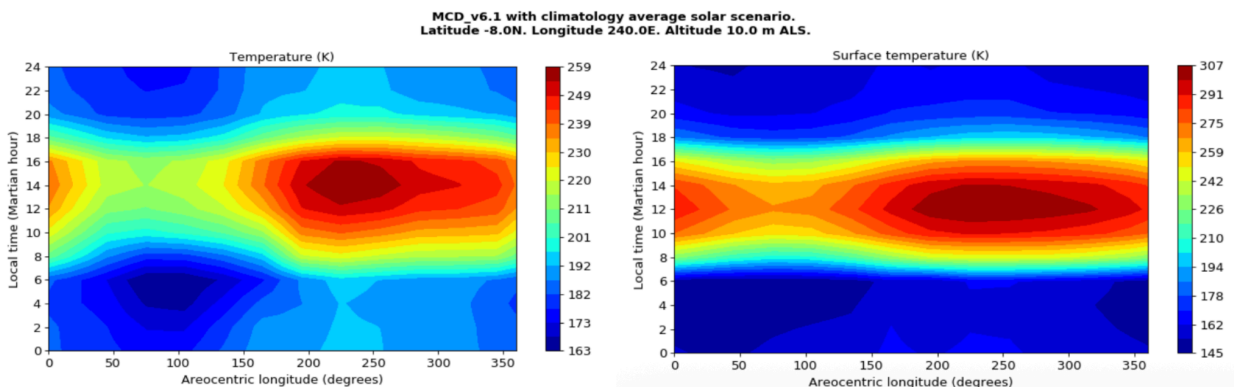
The heat distribution subassembly ensures that the heat generated by the electronics and heaters is equally distributed throughout the spacecraft. This unit comprises the light, very efficient thermal straps made of copper. These straps transfer heat from hotter areas to colder ones and eliminate the risk of cold spots from occurring in spots where temperatures can fall below the 20 °C mark. In addition, they also transfer the excess heat to the spacecraft's structural framework and to the radiator for removal during the periods of warm weather. Thermal straps are very efficient since they are passive and don't require power. This means they're safe from mechanical failure and provide continuous thermal equilibrium. Furthermore, they also add redundancy to the system since they provide the heat with multiple paths to travel along. Therefore, a failure of a single strap or blockage along the path will not isolate a component thermally and lead to a system failure.

The environmental-interface subassembly regulates the spacecraft's radiative and conductive interactions with the Martian environment. There is a white-painted radiator on the top surface, which serves as the main source of heat rejection and waste removal, especially in the cases of internal electronics that may generate significant waste energy. The rest of the external faces are covered with Multi-Layer Insulation (MLI), which significantly limits heat loss during the night and prevents unwanted heat gain during the day. MLI keeps the internally generated heat and lessens the heater duty cycle, whereas the radiator guarantees that the inside will not be warmer than 20 °C when exposed to solar radiation or continuous use of the electronics. T

These three subassemblies work closely to create predictable and regulated heat-flow channels throughout the system. From the inside, heat generated by avionics, heaters, and other electronics is spread over the structure with the help of thermal straps. After that, any waste energy or heat is sent to the radiator, from where it can be released to the environment safely. On the outside, some solar radiation reaches the spacecraft through the radiator surface, whereas its MLI-covered walls lessen the impact of atmospheric and ground-temperature variations. There are also some limited conductive and convective exchanges with the Martian environment happening at the wheels and structural contact points, but these are very small compared to radiative effects. In general, the system works to ensure that all internal components are kept with an operational temperature of 20°C.

2.1.4.1 Thermal Control Subsystem Requirements

To ensure the thermal control subsystem operates successfully, a list of requirements has been generated that the subsystem must achieve. TCS-01 defines the overall goal of the system, ensuring that it can accomplish its primary task. The air and ground temperature ranges chosen are based on the following data (Millour).



This data shows the air and surface temperatures in Mars over the course of the day and year at our chosen location, so the coldest and hottest temperatures were used to establish the range of temperatures in which the system must operate (TCS-01). A buffer of 10% has been added to those ranges to account for variations in location throughout the mission and to increase the overall factor of safety. TCS-02, TCS-06, and TCS-08 are based on system constraints, either from the mission or from other

subsystems, and they ensure that our system stays within mass, cost, size, and power constraints. TCS-07 also relates to the power constraint by requiring that some form of passive heating is used, reducing power usage while still meeting the heating needs of the system. Finally, TCS-03, TCS-04, and TCS-05 ensure that TCS-01 is met. A preliminary heat flow map showed that the system would lose heat to the environment, so heat generation is critical, and understanding the system's internal temperature at various locations ensures that the appropriate amount of heat is added to the system by the heat generation subassembly.

Req #	Requirement	Rationale	Parent Req.	Child Req.	Verification Method	Req Met?
TCS-01	The TCS shall maintain a consistent internal system temperature within component operational temperatures while exposed to ground temperatures within the range 128K - 324K and air temperatures in the range 149K - 272K	The system must remain within operational temperatures at all times, and this range was chosen based on data from the Mars Climate Database with a 10% added buffer to account for variance and other factors.	SYS-08	TCS-03 TCS-04	Test	Met
TCS-02	All purchased components, manufacturing, testing facilities, and other costs shall total less than TBD dollars	Overall cost must remain within budget, and TBD dollars have been allocated to the completion of the TCS.	MR-09		Inspection	Met
TCS-03	The TCS shall be capable of sensing internal temperature	Readings from temperature sensors inform the TCS how much heat to add to the system to maintain operational ranges for all subsystems	TCS-01	TCS-05	Inspection	Met
TCS-04	All heat generation hardware shall have at least one redundant component	Any component overheating or freezing would inhibit it from performing its intended function, so the consequences of a	TCS - 01		Inspection	Met

		heating component failing are significant.				
TCS-05	At least 3 temperature sensors shall be present in the system	Adds redundancy while showing where heat may need to be directed	TCS-03		Inspection	Met
TCS-06	Power consumption by the TCS shall not exceed 75 Watts	Power is needed for all operations, so maintaining TCS power consumption below a specified threshold allows other systems to utilize consistent power levels for the completion of their tasks	SYS-02	TCS-07	Inspection	Met
TCS-07	Passive heating techniques shall be utilized within the TCS	Passive heating techniques (such as insulation and coatings) do not take power, so using them will minimize overall power usage	TCS-06		Inspection	Met
TCS-08	The TCS shall minimize its mass and volume	The overall system has strict mass and stowed volume constraints, so minimizing TCS mass and volume can allow for more to be used elsewhere	MR - 05		Inspection	Met

2.1.4.2 Thermal Control Subsystem Recovery and Redundancy Plans

Recovery

The TCS components which will rely on power will be the heaters and temperature sensors. A heater failure will be identified through the corresponding temperature sensor not changing readings after the heater is turned on. If that occurs, the backup heater will immediately be activated. The primary heater will be turned off and after 10 minutes, it will be powered on again to test if it was a temporary or permanent failure. If temporary, the primary heater will continue its use, while if permanent, it will be fully powered off and the backup heater will take over the needed heating in that location. If

both heaters experience a total failure, they will each be retested once every 24 hours, but the heating needed will come from other heaters and the heat straps will be used to distribute that extra heat to the location which has lost its heating. A temperature sensor failure will be identified through large differences between it and other sensors, and if a failure occurs, a full reset of the sensor will take place and its readings will be reevaluated. If the readings remain significantly different from other sensors, its readings will be permanently ignored, and an estimation will be made for the accurate temperature at that location based on the gradient between nearby sensors.

Redundancy

Throughout the TCS, there will be multiple redundant systems which will help in case of failures or malfunctions. Each patch heater will have a backup, and “shortcut routes” for heat distribution will be added with the thermal straps, so no component or subsystem is completely reliant on one thermal strap. The extra patch heaters are true backups, and will only be put in use if the originals break, but the extra thermal straps will help distribute temperature more quickly even when all thermal straps are operational. For the environment interface, the only redundancy will be at connection points, ensuring that one faulty connection will not compromise the system. Finally, the temperature sensors will not have direct backups for each sensor, but using multiple of them ensures that one failure is not a significant problem.

2.1.4.3 Thermal Control Subsystem Manufacturing and Procurement Plans

All the components necessary for the Thermal Control system are common products in multiple applications, so there are some high quality off the shelf parts, which will be used for this mission for the simplicity of procurement and cheap cost (NASA).

Electric Resistance Patch Heaters are a common product in multiple applications, so there are some high quality off the shelf parts, which will be used for this mission for the simplicity of procurement and cheap cost (NASA). The primary supplier is Minco Products, from which 6 (3 primary, 3 redundant) of the HAP6948 All-Polyimide Thermofoil Heaters will be purchased. Each of these heaters has dimensions of 1.5in x 3.0in x 0.01in, provides a maximum power of 250 Watts, costs \$144.00, and has a standard lead time of 3 weeks if no heaters are already in stock (Minco). The backup supplier is DwyerOmega with the PLM-304/10 Polyimide Flexible Heater. This product has dimensions of 4in x 3.0in x 0.01in, provides a max power of 120 Watts, costs \$94.11, and has a standard lead time of 5 days if no heaters are in stock, and like the Minco heater, 6 of these would be included in the final subsystem if the primary option is unavailable (DwyerOmega). The Minco heater was chosen as the

primary option over the DwyerOmega heater because it provides more maximum power over less area, though both options will satisfy the requirements of the system.

Heat straps are not as widely used as patch heaters, so off the shelf parts are not as readily available (NASA). Instead, a contractor will be selected to make the desired heat straps. The primary contractor for this is Technology Applications Incorporated, specifically their CuTS (Copper Thermal Strap) line (Technology Applications). The backup contractor will be Thermal Management Technologies with their standard flexible thermal straps (Thermal Management Technologies). The estimated mass of this subsystem is 10kg, with cumulative length of roughly 3 meters to ensure the vehicle has consistent temperature throughout. This estimated mass was used in the mission concept cost estimate tool, leading to an overall cost estimate of just under \$2700. For either contractor, an expected lead time is 6 weeks (NASA).

Similar to the heat straps, the MLI will be made through a contractor. The primary option is Dunmore Aerospace, with their aluminized Kapton film (Dunmore). The backup is A&R Tarpaulins, which will be able to produce a similar product (A&R Tarpaulins). White paint can be bought commercially off the shelf and applied to the appropriate surface in-house. The primary supplier is AZ technologies, and the backup is Socomore (AZ technologies and Socomore). The total lead time for this subassembly is estimated to be 6 months due to the difficulty in production of MLI. The cost of these products are more difficult to estimate because no exact quotes are available and the mass is nearly negligible, so a very rough high-end estimate of the cost for this entire subassembly is \$15,000.

A table of suppliers, estimated cost, and lead time for each subassembly and the total Thermal Control System (TCS) is given below. The estimated cost and lead time for the complete TCS include a 50% increase from what is required for the subassemblies alone to include consideration of assembly and integration of the subassemblies (NASA).

	Primary Supplier	Backup Supplier	Estimated Cost	Lead Time
Patch Heaters	Minco Products	DwyerOmega	\$864	3 weeks
Thermal Straps	Technology Applications Incorporated	Thermal Management Technologies	\$2700	6 weeks
MLI and White Paint	Dunmore Aerospace (MLI), AZ Technologies (White Paint)	A&R Tarpaulins (MLI), Socomore (White Paint)	\$15,000	6 months

Complete TCS	-	-	\$27,846	9 months
--------------	---	---	----------	----------

2.1.4.4 Thermal Control Subsystem Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
TCS-01	The TCS shall maintain a consistent internal system temperature within component operational temperatures while exposed to ground temperatures within the range 128K - 324K and air temperatures in the range 149K - 272K	Test	Measurements need to be conducted under controlled environments utilizing instrumentation and environmental simulation to determine whether the system will operate under desired temperature ranges.	Conduct hot and cold testing on all temperature sensors and critical nodes in a thermal vacuum chamber. Verify temperature ranges fall within acceptable limits.
TCS-02	All purchased components, manufacturing, assembly and integration costs, testing equipment, and other costs shall total less than 10 million dollars	Inspection	Compliance can be verified through the review of documentations and invoices. No further testing, demonstrations or analysis is required	Inspect financial records, vendor quotes, and manufacturing costs. Compare the total cost with the TBD budget.
TCS-03	The TCS shall be capable of sensing internal temperature	Demonstration	Operate the part to determine whether it is functioning and well suited for the environment	Power on the system and demonstrate that temperature data is generated, readable, and accurate on the user interface module

TCS-04	All heat generation hardware shall have at least one redundant component	Inspection	Can be determined with a visual inspection of the system. No further testing, analysis, or demonstration is required	Inspect drawings, wiring diagrams, and hardware assembly to confirm the presence of redundant components
TCS-05	At least 3 temperature sensors shall be present in the system	Inspection	Can be determined with a visual inspection of the system. No further testing, analysis, or demonstration is required	Inspect CAD models, drawings, wiring diagrams, and hardware assembly to confirm the presence of redundant components
TCS-06	Power consumption by the TCS shall not exceed 75 Watts	Test	Requires the use of special equipment and instrumentation to evaluate power draw on the system	Measure the power draw using multimeters or other electronic testing equipment during TCS operations. Verify power draw does not exceed 75 watts
TCS-07	Passive heating techniques shall be utilized within the TCS	Inspection	Can be determined with a visual inspection of the system. No further testing, analysis, or demonstration is required	Inspect the insulations, coatings, and other thermal equipment. Verify that some form of passive thermal equipment are used
TCS-08	The TCS shall minimize its mass and volume	Analysis	Utilize analytical techniques and formulas to determine acceptable values	Utilize CAD software to determine the mass and volume of the system. Compare with system allocations and ensure it remains as

				minimal as possible
--	--	--	--	---------------------